Hayden Choi

haydenc9898@gmail.com | haydenchoi.com | linkedin.com/in/hayden-choi9 | github.com/Hayden9898

Education

McMaster University — Expected May 2028
Honours Computer Science Co-op, Bachelor of Applied Science (GPA: 3.9/4.0) — *Hamilton, ON*

- Relevant Coursework: Data Structures & Algorithms, Software Development, Distributed Databases, Operating Systems

Experience

Shopify — Jan 2027 – Apr 2027
Software Engineer Intern — *Toronto, ON*

- Incoming Winter 2027

Wealthsimple — Sept 2026 – Present
Software Engineer Intern — *Toronto, ON*

- Deployed a customer-facing risk event API in the **Rails** monolith, streaming device session signals through **Kafka** into the fraud decision platform for real-time evaluation, preventing **$1.3M+** in quarterly account takeover losses
- Integrated mobile telemetry in **Kotlin** emitting call and screen-capture flags synchronously at transaction submit across **4M+** client sessions, enforcing a fail-closed consumer contract that escalates dropped detections instead of passing them
- Shipped composite rules scoring device context against transaction behaviour, focal-blocking **3** money movement flows
- Deprecated a legacy bank linking decision service, removing **14** call sites reducing decision platform latency by **12%**

Trend Micro — May 2026 – Aug 2026
Software Engineer Intern — *Ottawa, ON*

- Built a behavioral ranking framework scoring **1,500+** risk event types on real fleet remediate-vs-dismiss outcomes, mining **8M+** human action decisions from **2,000+** enterprise tenants on a **Trino/Iceberg** lakehouse
- Designed a cross-repo **CI/CD** pipeline cutting the **5 day** risk event definition onboarding down to **2 days**
- Shipped a credential-free **EKS** service across **12** AWS regions on **Pod Identity**, owning its **Terraform** footprint
- Audited a pre-release **gRPC** connector SDK, uncovering **7+** packaging, versioning, and API-contract defects

Knockri — Dec 2025 – Apr 2026
Software Engineer Intern — *Toronto, ON*

- Created an AI agent to autonomously resolve support tickets, cutting inbox volume by **83%** and support costs by **61%**
- Reduced transcription tool latency **74%** by resolving **N+1 GraphQL** queries and refactoring data-fetching architecture
- Implemented a **Heartbeat API** using **GraphQL/UUID4** to restrict **7,000+** duplicate launches via backend locking

Brighter Signals BV — May 2025 – Aug 2025
Software Engineer Intern — *Markham, ON*

- Developed **5+** specialized prototypes for investor demos which secured **$1.6M+** in funding for **MVP** development
- Co-authored an **IROS 2025** paper and named a **Best Poster Finalist** for research on data-rich tactile sensors

Mana Immigration — May 2024 – Sept 2024
Software Engineer Intern — *Markham, ON*

- Architected **6+** full-stack Flutter apps for international clients using a Factory Pattern for reusable UI components
- Boosted app responsiveness by **50%** by implementing **Deferred Loading** for complex navigation modules

Projects

WatchTower | CSE Challenge Winner, Hack the North 2026 — Devpost | GitHub
FastAPI, React, TimescaleDB, scikit-learn, LLMs

- Built a hybrid detector pairing **6** rules with an **Isolation Forest** to correlate **180K+** HTTP logs into incidents

Proteus | 2nd Overall & Best Use of DigitalOcean, cuHacking7 — Devpost | GitHub
FastAPI, Next.js, MongoDB Atlas Vector Search, Redis, DigitalOcean

- Created a genomic IDE turning plain-English goals into scored DNA with predicted folds, accelerating wet lab testing

ColdApproach AI — GitHub
Next.js, FastAPI, AWS (SES, SQS, Lambda), OpenAI, Hunter.io

- Built an AI outreach platform with **FastAPI** and **Next.js** using **Hunter.io** discovery and a two-pass **GPT** pipeline

Technical Skills

Languages: Python, TypeScript, Ruby, Java, Kotlin, JavaScript, C++, SQL
Frameworks & Tools: Ruby on Rails, FastAPI, React, GraphQL, Temporal, Docker, Linux, Git, CI/CD
Data & Cloud: AWS (EKS, S3, Glue, Athena), Kubernetes, Terraform, Kafka, Trino, Iceberg, PySpark